



Can Fintech enhance firms' ambidextrous innovation capabilities? Perspective of organizational resilience

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ABSTRACT

This study examines how financial technology (Fintech) influences firms' ambidextrous innovation, focusing on breakthrough and incremental innovation, with organizational resilience as a moderating factor. The analysis uses a panel dataset of 23,007 observations from Chinese A-shared companies (2010–2022), addresses endogeneity with instrumental variables, propensity score matching, and the Heckman correction. Results indicate that Fintech significantly enhances innovation, with a greater impact on breakthrough innovation. Organizational resilience magnifies this effect, highlighting its critical role. Robustness tests confirm the findings. The study provides policymakers and managers with insights into leveraging Fintech and resilience to drive innovation in dynamic environments.

1. Introduction

The rapid advancement of financial technology (Fintech) is reshaping industries, competitiveness, and the way firms innovate. Fintech improves efficiency, financial accessibility, and technological adaptability by merging big data analytics, artificial intelligence (AI), and blockchain (Berman et al., 2022). Previous studies recognize Fintech's significance in decreasing financial limitations and facilitating knowledge flow (Chen et al., 2019); however, its direct impact on ambidextrous innovation—balancing breakthrough and incremental innovation—remains underexplored. This balance is essential for continuing competitiveness since firms operate in increasingly volatile markets. Fintech promotes drastic and ongoing improvements; however, its mechanics are unknown.

Research suggests that Fintech accelerates real-time data processing, lowers financial obstacles, and promotes disruptive innovation (Shi and Lu, 2024). Additionally, it streamlines operations, improves iterative design processes, and enhances product refinement (Chen, 2024). However, empirical studies on how Fintech helps firms pursue both innovation types are sparse. Organizational resilience—a firm's ability to adapt, recover, and thrive in difficult environments—also remains underexplored (Wang et al., 2023; Xue and Wang, 2024; Xu et al., 2024). Resilient firms may be better able to absorb technology upheavals and use Fintech's innovative potential. There is a significant gap in the literature regarding how resilience modifies the Fintech–innovation link.

This study addresses two crucial research questions: (1) how does Fintech affect firms' ambidextrous innovation? (2) To what extent does organizational resilience moderate this relationship? Answering these questions advances Fintech and innovation scholarship in three ways. First, it adds Fintech and resilience to the ambidextrous innovation framework, offering a new perspective on how firms use digital technologies to innovate. Second, it uses instrumental factors, propensity score matching (PSM), and the

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Heckman adjustment to address endogeneity in a panel dataset of 23,007 observations from Chinese A-shared companies from 2010 to 2022. Third, it examines ownership structure and industry pollution heterogeneity to determine how firm characteristics affect Fintech–innovation.

This study has important practical implications in addition to its theoretical significance. Findings instruct policymakers on Fintech ecosystem development, regulatory reform, and digital transformation strategies to boost national innovation capability. Results emphasize the need of combining Fintech adoption with resilience-enhancing initiatives for business leaders, enabling them to reduce risks, improve innovation, and maintain long-term competitive advantages. This research addresses these key challenges and contributes to the conversation on digital finance, organizational adaptation, and technology-driven innovation in a fast-changing economy.

2. Literature review

2.1. Fintech and innovation

Fintech integrates modern digital technologies into financial services to improve efficiency, accessibility, and innovation. Fintech comprises blockchain, AI, machine learning, big data analytics, and digital payment platforms (Berman et al., 2022). Fintech improves financial transactions and increases corporate agility using these technologies. It helps firms innovate by lowering financial obstacles, improving resource allocation, and enabling quick experimentation.

The relationship between Fintech and innovation is based on its capacity to improve financial access, knowledge dissemination, and decision-making processes. Fintech provides real-time data analytics, predictive modeling, and algorithmic forecasting to help firms identify emerging opportunities and invest in high-risk, high-reward innovations (Chen et al., 2019). The availability of alternative financing mechanisms, such as crowdfunding, decentralized finance, and venture capital platforms powered by Fintech, further supports firms in pursuing ambitious R&D projects that push technological boundaries.

Fintech's capacity to improve operational efficiency and iterative development cycles also benefits incremental innovation, which focuses on improving existing products and processes. As firms refine their products and services with digital twins, process automation, and cloud-based modeling systems, utility model patents and design improvements also increase (Shi and Lu, 2024). Using Fintech-driven predictive analytics and customer behavior data, firms may customize product additions to market demand, maintaining competitiveness.

Fintech accelerates radical technological advances and facilitates continual improvements, confirming its central role in modern innovation ecosystems. Understanding its implications for ambidextrous innovation—balancing exploration and exploitation—becomes essential for researchers and practitioners as firms increasingly employ Fintech-enabled tools and platforms.

2.2. The role of organizational resilience

A firm's resilience is its ability to survive, adapt, and recover from disruptions while pursuing strategic goals (Wang et al., 2023; Xue and Wang, 2024). This dynamic skill is increasing as important for firms to maintain operational continuity and drive innovation in volatile situations. Resilience allows firms to adapt, adjust, and seize opportunities, facilitating reactive and proactive crisis responses (De Carvalho et al., 2016). Firms with robust resilience frameworks are better able to overcome uncertainty and maintain competitive advantage in volatile business environments.

A critical aspect of resilience is its relationship with innovation. Resilient organizations often leverage adversity as a catalyst for innovation, transforming challenges into opportunities for growth and development. For instance, green innovation boosts resilience by linking organizational goals with sustainable practices and external acknowledgment (Dai, 2025; Khan et al., 2023). Innovative responses to crises such as the COVID-19 pandemic have helped firms survive and grow in disrupted marketplaces (Vargas-Hernández, 2022). These data show that resilience is a foundation and result of innovative solutions.

Organizational resilience helps integrate innovative technology and systems. Resilient companies are better at digital transformation, using AI and IoT. This integration helps them adapt to market changes and customer needs (Gao et al., 2024; Garrido-Moreno et al., 2024). Beyond technological adoption, resilient firms are more likely to promote a culture of continuous learning and adaptability, which sustains innovation (Buliga et al., 2016; Dai et al., 2025).

Resilience mediates firms' innovative work behavior. Staff may experiment, take risks, and execute new ideas with resilience's supporting settings and adaptable capacities. This mediating role emphasizes the importance of resilience in linking organizational support structures with innovation outcomes (Zhou et al., 2023). These findings demonstrate resilience's transformative role in helping firms stay innovative and competitive in the face of adversity, making it a key facilitator in modern organizational strategies.

2.3. Hypothesis development

Drawing on the preceding discussions, three hypotheses are developed to lead the empirical analysis:

H1. Fintech positively influences breakthrough innovation.

The use of Fintech provides firms with innovative tools for ideation, experimentation, and patent development. Advanced technologies like AI and blockchain make it easier to develop ground-breaking solutions that align with emerging market demands.

H2. Fintech positively influences incremental innovation.

Fintech's data-driven solutions allow firms to optimize their present goods and processes, enabling continual improvement and refinement. These advancements result in consistent generation of utility models and design improvements, emphasizing Fintech's role in incremental innovation (Berman et al., 2022).

H3. Organizational resilience positively moderates the relationship between Fintech and innovation (breakthrough and incremental).

Firms with high resilience are better positioned to capitalize on Fintech's potential, transforming technological advancements into innovative outcomes. Resilience reduces the risks associated with digital adoption, allowing firms to adapt flexibly and proactively to market problems (Frare et al., 2023).

3. Methodology

3.1. Data source and sample

The analysis utilizes panel data from Chinese A-shared companies from 2010 to 2022. This dataset contains essential measures on Fintech development, organizational resilience, and innovation, with a sample size of 23,007 observations. Data sources include government reports, patent databases, financial information, and the Wind and CSMAR databases. The dataset was meticulously filtered to assure quality, completeness, and reliability, with outliers and missing data points resolved using robust statistical approaches.

3.2. Model specification

The empirical analysis uses panel regression models to examine the association between Fintech development and firm ambidextrous innovation, with organizational resilience as a moderating variable. The baseline and moderating models are as follows:

3.2.1. Baseline model

The baseline model assesses the direct impact of Fintech on breakthrough innovation ($Innovation_{brt}$) and incremental innovation ($Innovation_{icm}$):

$$Y_{it} = \beta_0 + \beta_1 FinTech_{it} + \sum_{k=1}^n \beta_k CV_{it} + \varepsilon_{it},$$

where Y_{it} denotes the dependent variable for firm i in year t . $FinTech_{it}$ represents the Fintech index for the city in which the firm operates. $\sum_{k=1}^n \beta_k CV_{it} + \varepsilon_{it}$ captures the control variables. ε_{it} is the error term. This approach determines whether Fintech development has a statistically significant association to firms' breakthrough and incremental innovation.

Next, to evaluate the moderating effect of organizational resilience (OR_{it}), the interaction term ($FinTech_{it} \times OR_{it}$) is included:

$$Y_{it} = \beta_0 + \beta_1 FinTech_{it} + \beta_2 OR_{it} + \beta_3 (FinTech_{it} \times OR_{it}) + \sum_{k=1}^n \beta_k CV_{it} + \varepsilon_{it},$$

where OR_{it} represents organizational resilience for firm i in year t . $FinTech_{it} \times OR_{it}$ captures the interaction effect of Fintech development and organizational resilience. All other terms follow the baseline model's definitions. The coefficient β_3 on the interaction term reflects whether organizational resilience improves or diminishes Fintech's impact on innovation results. A positive β_3 suggests that resilience increases Fintech's effect on innovation, whereas a negative β_3 indicates a dampening effect.

3.3. Variable measurement

This study examines two aspects of corporate innovation capability. The first dimension assesses breakthrough innovation ($Innovation_{brt}$) by using the natural logarithm of the number of invention patents awarded independently in the current year plus 1. The second dimension measures incremental innovation ($Innovation_{icm}$) by calculating the natural logarithm of the sum of utility models and design patents plus one.

The independent variable, Fintech development ($FinTech$), is measured using a city-level Fintech index. This index was constructed by finding 48 Fintech-related keywords and using web crawler technology to match them with prefecture-level cities. The logarithmic transformation of keyword match results measures Fintech development. To provide robustness, the number of Fintech firms in a city ($FinTech_{replace}$) is used as an alternative proxy.

The computation for organizational resilience (OR_{it}) follows the expression

$$OR_{it} = \frac{(\Delta SaleAve_{t-1} / \Delta SaleAve_t)}{(\Delta AllSaleAve_{t-1} / \Delta AllSaleAve_t)},$$

where $SaleAve$ denotes the enterprise's sales avenue in the current year, and $AllSaleAve$ denotes the sales avenue of all enterprises under

consideration. A positive OR_{it} value indicates superior organizational resilience, whereas a negative value suggests lower resilience compared with industry peers.

The research incorporates several control variables to account for firm-specific characteristics that might influence innovation capabilities. Enterprise size ($Size_{it}$) is measured by the natural logarithm of total assets. Return on equity (ROE_{it}) is calculated as net profit divided by average shareholders' equity. Meanwhile, financial leverage (FL_{it}) is computed as the ratio of earnings before interest and tax to earnings before tax. Board size ($Board_{it}$) is measured as the natural logarithm of the number of board members. Enterprise value (TQ_{it}) is assessed through Tobin's Q, calculated as the sum of market value of outstanding shares, nonoutstanding shares multiplied by net assets per share, and book value of liabilities, divided by total assets.

This methodological framework combines rigorous data collection, well-defined variable measurements, and robust statistical techniques to explore the interplay between Fintech, organizational resilience, and ambidextrous innovation. This study provides a reliable foundation for uncovering the mechanisms driving innovation in a dynamic environment by leveraging a comprehensive dataset and addressing potential biases through robustness checks.

4. Results and discussion

4.1. Summary statistics

Table 1 presents a descriptive overview of the key variables included in the analysis. The dependent variables, breakthrough innovation ($Innovation_{brt}$) and incremental innovation ($Innovation_{icm}$), exhibit a mean of 0.717 and 1.150, respectively. The median (p50) values of these variables suggest a concentration around lower innovation outputs, indicating skewness in the distribution, particularly for $Innovation_{brt}$, where half of the observations record zero patent counts.

The independent variable, Fintech ($FinTech$), has a mean value of 4.996 and a standard deviation of 1.589, indicating significant diversity in Fintech development levels between prefecture-level cities. The moderator, organizational resilience (OR), has an average of 1.464 but is highly variable, with values ranging from −11.04 to 31.42. This substantial range highlights the variations in the adaptive capacity of firms in overcoming external problems.

Control variables also have varied distributions. The average firm size ($Size$) is 22.27, whereas the average return on equity (ROE) and financial leverage (FL) are 0.0700 and 1.236, respectively. The board size ($Board$) is fairly consistent, with a mean of 2.123, whereas Tobin's Q ($TobinQ$), a proxy for firm value, has a mean of 2.014, indicating reasonable market valuation levels in relation to total assets.

4.2. Baseline regression analysis

The data in Table 2 show that Fintech firms boost breakthrough and incremental innovation. Fintech has a high positive effect on breakthrough innovation in Model 1, validating H1. The impact on incremental innovation (Model 2) is likewise significant, supporting H2. The larger impact on breakthrough innovation suggests that Fintech largely promotes access to cutting-edge tools and technology essential for radical, high-impact breakthroughs. Simultaneously, its role in incremental innovation is evident, as it fosters efficiency improvements and optimizes iterative product development.

Several firm-specific characteristics affect innovation. Firm size positively affects breakthrough and incremental innovation, demonstrating the role of resources in technological growth. ROE strongly correlates with incremental innovation, suggesting that firms with greater financial flexibility can sustain and improve current capabilities. Board size also positively affects breakthrough innovation, showing that robust governance structures encourage strategic R&D investment decision-making and risk-taking. However, FL does not significantly impact either category of innovation, suggesting that debt levels alone do not determine firms' inventive capacities.

These results match previous studies. According to Berman et al. (2022), Fintech helps firms innovate by lowering financial obstacles and improving resource allocation. Additionally, Lee et al. (2019) suggest that Fintech reduces information asymmetry and financing constraints, helping high-risk, high-reward breakthrough innovation firms. The findings support the theory that efficient resource management, governance quality, and strategic financial tools are required for firms to use Fintech to drive exploratory and exploitative innovation.

Table 1
Summary statistics.

Variable	N	Mean	p50	SD	Min	Max
Innovation_brt	23,007	0.717	0	1.006	0	4.654
Innovation_icm	23,007	1.150	0.693	1.376	0	5.591
FinTech	23,007	4.996	5.247	1.589	1.099	7.854
OR	23,007	1.464	0.840	4.676	−11.04	31.42
Size	23,007	22.27	22.07	1.279	20.12	26.11
ROE	23,007	0.0700	0.0740	0.104	−0.416	0.344
FL	23,007	1.236	1.044	0.772	−0.382	5.784
Board	23,007	2.123	2.197	0.197	1.609	2.639
TobinQ	23,007	2.014	1.639	1.174	0.844	7.301

Table 2
Baseline regression results.

Variables	(1) Innovation_brt	(2) Innovation_icm
FinTech	0.055*** (6.670)	0.046*** (4.435)
Size	0.158*** (25.831)	0.042*** (5.504)
ROE	0.413*** (6.801)	1.385*** (18.103)
FL	−0.012 (−1.563)	0.001 (0.122)
Board	0.133*** (4.053)	0.347*** (8.423)
TobinQ	0.055*** (8.819)	0.003 (0.432)
Constant	−4.079*** (−23.540)	−2.086*** (−9.563)
Observations	23,007	23,007
R-squared	0.221	0.340
IND	FE	FE
YEAR	FE	FE

4.3. Moderation effect analysis

The moderating effect results in Table 3 demonstrate that OR significantly enhances the positive impact of Fintech on breakthrough and incremental innovation. The interaction term between resilience and Fintech is statistically significant in both models for breakthrough innovation (Model 2) and incremental innovation (Model 4), supporting H3. These findings suggest that resilient firms can better use Fintech technologies to innovate. Resilience increases Fintech's impact by allowing firms to adapt to technology changes and manage external obstacles. The slightly stronger moderating effect observed for incremental innovation suggests that resilience is crucial to optimizing current processes and capacities.

The direct effect of Fintech on innovation remains consistently significant across all models, highlighting its position as a fundamental enabler of breakthrough and incremental innovation. Notably, OR alone does not directly impact innovation outcomes, suggesting its influence is most significant when paired with Fintech. This emphasizes the importance of an integrated approach where resilience supports the effective use of digital tools to foster innovation.

These findings support Fintech-organizational adaptability studies. Lee et al. (2019) noted Fintech's capacity to reduce information asymmetry and increase financial and operational flexibility, whereas Buliga et al. (2016) stressed resilience's importance in helping firms flourish in complicated contexts and uncertainty. These findings highlight the synergistic relationship between Fintech and resilience, which improves a firm's ability to innovate effectively in unpredictable situations.

Table 3
Moderating effect.

Variables	(1) Innovation_brt	(2) Innovation_brt	(3) Innovation_icm	(4) Innovation_icm
FinTech	0.055*** (6.673)	0.056*** (6.790)	0.046*** (4.436)	0.048*** (4.582)
OR × FinTech		0.003*** (2.588)		0.004*** (3.151)
OR	−0.002 (−1.480)	−0.001 (−0.689)	−0.001 (−0.683)	0.000 (0.235)
Size	0.158*** (25.855)	0.158*** (25.811)	0.042*** (5.515)	0.042*** (5.462)
ROE	0.435*** (6.957)	0.416*** (6.596)	1.397*** (17.737)	1.367*** (17.232)
FL	−0.013 (−1.578)	−0.013* (−1.655)	0.001 (0.114)	0.000 (0.021)
Board	0.132*** (4.049)	0.133*** (4.065)	0.347*** (8.421)	0.348*** (8.442)
TobinQ	0.055*** (8.862)	0.054*** (8.711)	0.004 (0.454)	0.002 (0.283)
Constant	−4.083*** (−23.564)	−4.085*** (−23.576)	−2.089*** (−9.574)	−2.092*** (−9.588)
Observations	23,007	23,007	23,007	23,007
R-squared	0.2207	0.221	0.3395	0.340
IND	FE	FE	FE	FE
YEAR	FE	FE	FE	FE

Implications for policymakers are clear: supporting innovation ecosystems requires increasing Fintech uptake and OR. Investments in digital infrastructure and capacity-building programs that improve firm adaptability can boost innovation. For managers, building flexible operations, robust governance, and strategic firm management systems can help firms maximize on Fintech's innovation potential.

4.4. Addressing endogeneity issues

The results in Table 4 address endogeneity problems using an instrumental variable method, in which the lag value of Fintech is used as an instrument. This method reduces the risk of reverse causality and variable bias, ensuring robust conclusions. The two-stage regression reveals the significant positive link between Fintech and breakthrough and incremental innovation, supporting H1 and H2.

The lagged Fintech variable is highly significant in Models 1 and 3, with an R^2 of 0.962. A significant correlation between the instrument and the endogenous independent variable meets the relevance criteria for instrumental variables. Lag Fintech's consistent performance across breakthrough and incremental innovation models supports its legitimacy as an instrument.

Models 2 and 4 show that the expected Fintech variable still positively and significantly affects innovation results. These results show the causal impact of Fintech on innovation. Fintech provides access to innovative technologies for breakthrough innovation in products and processes. Fintech drives efficiency improvements and iterative developments for incremental innovation, a lesser but still significant effect. Findings support Fintech's function in resource allocation and information dissemination.

Control variables mostly behave as expected. Firm size positively correlates with innovation, supporting the resource-based idea that larger firms are better equipped to innovate. Profitability significantly impacts both forms of innovation, especially incremental innovation, underscoring the importance of financial flexibility. Board size contributes positively to innovation, especially incremental advances, showing the importance of governance in adaptive methods. Insignificant FL suggests debt levels do not encourage innovation.

Table 5 results use PSM and Heckman adjustment method. Fintech positively and significantly affects breakthrough and incremental innovation in PSM Models 1 and 2. Even after matching firms with identical characteristics, Fintech adopters outperform others in innovation. The slightly stronger effect on breakthrough innovation suggests that Fintech adoption fosters high-impact, new inventions. Fintech improves processes and products, boosting incremental innovation.

The Heckman correction method (Models 3 and 4) accounts for selection bias and supports these findings. The significance of the inverse Mills ratios (imr1 and imr2) in both models highlights the importance of adjusting for sample selection concerns. After controlling for selection bias, Fintech coefficients are higher than PSM models, suggesting that unobserved factors may have overestimated its impact. The data indicate Fintech's continuous influence on both types of innovation.

Overall, the Heckman correction, instrumental variables, and PSM endogeneity analysis provide robust evidence that Fintech causes innovation. By addressing potential reverse causality, omitted variable bias, and sample selection concerns, these methods strengthen the hypothesis that Fintech generates breakthrough and incremental innovation.

4.5. Additional robustness check

The results in Table 6 show additional robustness checks that support the original find. These checks include using an alternative independent variable ($FinTech_{replace}$), changing model specifications, and eliminating COVID-19 samples. Across all studies, Fintech

Table 4
Addressing endogeneity issues: instrumental variable.

Variables	(1) FinTech	(2) Innovation_brt	(3) FinTech	(4) Innovation_icm
L.FinTech	0.911*** (257.378)		0.911*** (257.378)	
FinTech		0.068*** (7.177)		0.048*** (3.928)
Size	0.013*** (6.682)	0.150*** (18.571)	0.013*** (6.682)	0.035*** (3.769)
ROE	−0.022 (−1.130)	0.405*** (6.558)	−0.022 (−1.130)	1.302*** (16.037)
FL	−0.005 (−1.597)	−0.013 (−1.618)	−0.005 (−1.597)	−0.002 (−0.202)
Board	−0.029** (−2.573)	0.138*** (3.770)	−0.029** (−2.573)	0.355*** (7.933)
TobinQ	0.005** (2.324)	0.057*** (7.594)	0.005** (2.324)	0.012 (1.401)
Constant	0.158*** (2.627)	−3.780*** (−18.757)	0.158*** (2.627)	−1.149*** (−4.840)
Observations	19,382	19,382	19,382	19,382
R-squared	0.962	0.223	0.962	0.343
IND	FE	FE	FE	FE
YEAR	FE	FE	FE	FE

Table 5
Addressing endogeneity issues: PSM and Heckman.

Variables	(1) PSM		(2) Heckman	
	Innovation_brt	Innovation_icm	Innovation_brt	Innovation_icm
FinTech	0.056*** (4.916)	0.052*** (3.574)	0.299*** (3.266)	0.241*** (3.031)
imr1			3.208*** (2.674)	
imr2				1.406** (2.472)
Size	0.148*** (17.163)	0.027** (2.471)	0.100*** (4.409)	−0.105* (−1.747)
ROE	0.368*** (4.152)	1.505*** (13.399)	2.328*** (3.239)	2.694*** (5.034)
FL	−0.022* (−1.934)	−0.011 (−0.754)	−0.048*** (−3.097)	0.030* (1.950)
Board	0.132*** (2.892)	0.390*** (6.747)	0.015 (0.265)	0.349*** (8.477)
TobinQ	0.047*** (5.572)	−0.006 (−0.549)	0.245*** (3.427)	−0.025* (−1.811)
Constant	−3.792*** (−16.009)	−1.824*** (−6.076)	−7.821*** (−5.546)	−1.251*** (−3.113)
Observations	11,800	11,800	23,007	23,007
R-squared	0.214	0.345	0.221	0.340
IND	FE	FE	FE	FE
YEAR	FE	FE	FE	FE

consistently demonstrates a positive and robust impact on breakthrough and incremental innovation, validating past conclusions.

In Models 1 and 2, the alternative independent variable, $FinTech_{replace}$ (measured as the number of Fintech firms), is positively and significantly associated with breakthrough and incremental innovation. These results are congruent with the primary analysis, showing that Fintech proxies find consistent results. The significance of $FinTech_{replace}$ highlights that Fintech firms and the Fintech development index are strong predictors of innovation.

The alternate model specification confirms the significant positive association between Fintech and innovation in Models 3 and 4. The increased coefficients in these models suggest that Fintech's involvement in innovation may be considerably larger. The findings are not dependent on a specific analytical approach because the results are robust to model structure alterations.

Models 5 and 6, which remove data from the COVID-19 era, show that the association between Fintech and innovation remains significant even after accounting for the pandemic's specific disruptions. These results show that Fintech's impact on innovation is not only a pandemic-induced effect but a systemic effect across markets.

These findings' robustness check consistency supports the primary results' reliability and validity. This durability shows the strategic importance of Fintech development to promote innovation ecosystems and maintain competitive advantages across economic

Table 6
Additional robustness checks.

Variables	(1)		(2)		(3)	
	Innovation_brt	Innovation_icm	Innovation_brt	Innovation_icm	Innovation_brt	Innovation_icm
FinTech_replace	0.027*** (7.500)	0.012*** (2.685)				
FinTech			0.093*** (21.418)	0.239*** (40.932)	0.057*** (6.008)	0.042*** (3.607)
Size	0.158*** (25.879)	0.045*** (5.794)	0.042*** (7.045)	−0.097*** (−12.066)	0.124*** (16.747)	0.026*** (2.897)
ROE	0.416*** (6.832)	1.374*** (17.916)	0.592*** (9.028)	1.617*** (18.372)	0.369*** (4.517)	1.205*** (11.997)
FL	−0.013 (−1.641)	−0.002 (−0.192)	−0.015* (−1.783)	0.017 (1.485)	−0.011 (−1.229)	0.017 (1.521)
Board	0.133*** (4.051)	0.335*** (8.117)	−0.036 (−1.017)	0.045 (0.959)	0.155*** (4.057)	0.349*** (7.445)
TobinQ	0.055*** (8.796)	0.004 (0.472)	0.096*** (15.727)	−0.014* (−1.718)	0.027*** (3.446)	−0.013 (−1.372)
Constant	−4.043*** (−23.319)	−2.039*** (−9.337)	−0.829*** (−6.082)	1.912*** (10.452)	−3.363*** (−16.680)	−1.687*** (−6.807)
Observations	22,916	22,916	23,007	23,007	15,408	15,408
R-squared	0.222	0.339	0.040	0.077	0.220	0.311
IND	FE	FE	FE	FE	FE	FE
YEAR	FE	FE	FE	FE	FE	FE

and industry contexts.

4.6. Heterogeneity analysis

4.6.1. Ownership structure: state-owned versus nonstate-owned enterprises

The results in Panel A, Table 7 show that Fintech significantly boosts breakthrough innovation in nonstate-owned enterprises (non-SOEs), with a highly significant and positive coefficient ($\beta = 0.088, p < 0.01$). However, it has negligible impact on incremental innovation in these firms ($\beta = 0.001, p > 0.1$). This suggests that privately owned enterprises use Fintech to pursue disruptive innovations rather than process improvements. By contrast, SOEs likewise experience a significant positive effect of Fintech on breakthrough innovation ($\beta = 0.093, p < 0.01$); however, its impact on incremental innovation is negative and statistically negligible ($\beta = -0.024, p > 0.1$). This suggests that SOEs may struggle to sustain process improvements despite technological advances.

Firm size affects both forms of innovation, with a higher effect observed in non-SOEs. In the private sector, where strategic flexibility facilitates faster adoption of new technologies, larger enterprises can better deploy resources to technological improvements. ROE impacts incremental innovation more under both ownership arrangements, whereas SOEs have a weaker link. State-controlled firms frequently rely on government assistance rather than financial performance, therefore financial profitability may not be as important.

With breakthrough and incremental innovation in SOEs, board size is significant, showing that governance structures shape strategic innovation decisions in state-controlled firms. By contrast, non-SOE board size mostly affects incremental innovation.

Table 7
Heterogeneity check.

Panel A	Ownership structure			
Groups Variables	Non-SOE Innovation_brt	SOE Innovation_icm	Non-SOE Innovation_brt	SOE Innovation_icm
FinTech	0.088*** (8.401)	0.001 (0.058)	0.093*** (6.947)	-0.024 (-1.420)
Size	0.167*** (19.312)	0.150*** (15.111)	0.084*** (7.527)	0.053*** (4.385)
ROE	0.476*** (6.485)	0.211* (1.882)	1.394*** (14.781)	0.924*** (6.734)
FL	-0.052*** (-4.449)	0.021* (1.947)	-0.051*** (-3.427)	0.037*** (2.786)
Board	0.040 (0.966)	0.262*** (4.745)	0.205*** (3.842)	0.673*** (9.935)
TobinQ	0.058*** (7.975)	0.040*** (3.194)	0.008 (0.861)	0.016 (1.015)
Constant	-4.020*** (-16.924)	-4.248*** (-14.958)	-2.659*** (-8.718)	-3.024*** (-8.693)
Observations	14,355	8164	14,355	8164
R-squared	0.179	0.305	0.340	0.337
IND	FE	FE	FE	FE
YEAR	FE	FE	FE	FE
Panel B	Heavy polluted industry			
Groups Variables	Nonpolluted Innovation_brt	Polluted Innovation_icm	Nonpolluted Innovation_brt	Polluted Innovation_icm
FinTech	0.059*** (6.010)	0.033** (2.288)	0.054*** (4.404)	0.017 (0.847)
Size	0.180*** (25.589)	0.072*** (6.018)	0.045*** (5.219)	0.026 (1.550)
ROE	0.460*** (6.627)	0.462*** (3.789)	1.486*** (17.256)	1.184*** (7.048)
FL	-0.008 (-0.844)	-0.030** (-2.436)	-0.006 (-0.498)	0.000 (0.010)
Board	0.157*** (4.201)	0.039 (0.604)	0.330*** (7.114)	0.424*** (4.799)
TobinQ	0.059*** (8.544)	0.031** (2.221)	0.004 (0.428)	0.001 (0.045)
Constant	-4.653*** (-24.147)	-1.644*** (-5.382)	-2.134*** (-8.915)	-1.261*** (-2.994)
Observations	18,459	4548	18,459	4548
R-squared	0.240	0.126	0.365	0.154
IND	FE	FE	FE	FE
YEAR	FE	FE	FE	FE

Note: t-statistics in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Financial leverage (FL) negatively affects breakthrough innovation in SOEs and non-SOEs but positively impacts incremental innovation. Debt financing operational improvements rather than high-risk innovative projects may explain this.

These data show how different ownership configurations affect how firms utilize Fintech. Non-SOEs with strategic flexibility and efficiency-driven innovation cultures gain more from Fintech in developing technological innovations. SOEs, despite having financial technology, appear hampered by bureaucratic process. This suggests that policy interventions should focus on encouraging SOEs to incorporate Fintech-driven process innovations into their innovation initiatives to reduce inefficiencies and promote long-term technological progress.

4.6.2. Industry-specific effects: polluted versus nonpolluted industries

According to the results in Panel B, Table 7, Fintech significantly leads to breakthrough innovation in nonpolluted industries ($\beta = 0.059, p < 0.01$) compared with pollution-heavy industries ($\beta = 0.054, p < 0.01$). This suggests that cleaner industry firms are better positioned to benefit on technological advances owing to less governmental restraints and a focus on market-driven innovation. Fintech positively impacts incremental innovation in nonpolluted industries ($\beta = 0.033, p < 0.05$) but has a modest impact on polluted industries ($\beta = 0.017, p > 0.1$). This disparity highlights the challenges ecologically intensive firms experience, where compliance costs and regulatory requirements may detract from innovation-focused projects.

Firm size significantly affects nonpolluted industries, where larger firms appear to have well-established structures that enable Fintech-driven tactics. Pollution-heavy sectors may have operational restrictions related to regulatory compliance that weaken the effect of firm size on incremental innovation. Financial leverage significantly affects breakthrough innovation in polluted industries, indicating the difficulty of allocating financial resources to high-risk, high-reward innovation in heavily regulated areas.

Board size is associated with breakthrough innovation in nonpolluted industries, highlighting the importance of governance frameworks in long-term strategic innovation. In pollution-heavy businesses, the effect is inconsistent, suggesting that external constraints and environmental policies may shape innovation tactics more than internal governance measures.

These findings suggest that firms in nonpolluted industries are better positioned to use Fintech for breakthrough and incremental innovation owing to less regulatory constraints, greater financial flexibility, and a stronger focus on long-term market growth. By contrast, pollution-heavy firms confront structural barriers to Fintech integration into their innovation strategy. Policy initiatives can encourage green finance solutions and technological investments corresponding to environmental sustainability goals to maximize Fintech's benefits and allow pollution-intensive firms to innovate while following regulatory obligations.

These heterogeneity factors help the study understand how Fintech boosts creativity in context. Findings suggest that ownership structures and industry characteristics significantly impact how firms benefit on digital innovations, emphasizing the need for specific governmental initiatives to encourage innovation-driven growth.

5. Conclusions

This study provides compelling empirical evidence that Fintech helps firms innovate ambidextrously, with a greater impact on breakthrough innovation than incremental innovation. Through a robust methodological framework that includes instrumental variables, PSM, and the Heckman correction, findings show that OR significantly strengthens the Fintech–innovation relationship, highlighting its importance in fostering adaptive capabilities. Owing to structural and regulatory factors, Fintech favors non-SOEs and firms in nonpolluted industries more, according to the heterogeneity analysis.

These observations have substantial implications for legislators, managers, and industry leaders. Governments should develop targeted policies that assist digital transformation and improve Fintech infrastructure, especially in SOEs and pollution-intensive industries where innovation potential is still untapped. Business executives must combine Fintech-driven strategies with resilience-building techniques to keep firms competitive in increasingly uncertain conditions. Optimizing Fintech's innovation-enhancing potential requires strengthening governance structures, financial flexibility, and strategic agility.

This study adds resilience as a moderating factor to Fintech and organizational innovation theory beyond its practical contributions. It bridges gaps in understanding how digital finance accelerates radical and incremental innovation, providing a holistic perspective beyond traditional innovation frameworks.

Despite its contributions, the study has limitations that allow for further investigation. The long-term impact of Fintech on firms' innovation cycles, industry-specific digital regulations, and cross-country comparative studies could help deepen our understanding of Fintech's revolutionary potential. Future studies can build on this foundation to gain insights into technology, resilience, and corporate innovation initiatives.

CRedit authorship contribution statement

Yuanyang Liu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Xuemei Tian:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Yang Long:** Writing – review & editing, Validation, Supervision, Resources, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

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